StockSense AI

Student testing manual: install it, test every function, rate it

FOR	YOUR COMPUTER	TIME
Student testers and evaluators	Windows 10 (22H2) or 11, 64-bit	About 1 hour to set up, 3 to 4 hours to test

YOU HAND IN
Test results, rating sheet, bug reports

StockSense AI is a web app that helps a small shop decide **what to reorder, how much and when**. It forecasts sales with a machine-learning model (XGBoost) and turns the forecast into replenishment advice. It only advises: it never places orders. You will run the whole system on your own computer and check, one function at a time, that it does what it should.

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How to use this manual

Work through it in order. **Part A** gets the app running on your computer. **Part B** is the test cases: each one tells you who to sign in as, what to do, and what should happen. **Part C** is where you rate the app and report problems.

Reading a test case

AUTH-01 **The test's name: what it checks** — Who to sign in as

Before you start: anything that must already be true.

1. Numbered steps. Do exactly these, in order.

EXPECTED RESULT
What you should see. If you see this, the test passes.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks _____

- **Pass**: what happened matches the expected result.
- **Fail**: something different happened. Fill in a bug report and write its number in Remarks.
- **Blocked**: you could not do the test (for example an earlier step failed). Say why in Remarks.
- You can tick the boxes and type remarks in the HTML version, then print it to PDF with `Ctrl` + `P`. Ticks typed into the page are not saved if you close it, so print before you close.

i **The demo shop.** The app comes with a made-up shop: 50 products in 5 categories (food and beverages, personal care, household and cleaning, school and office supplies, hardware) and about two years of sales. No real customer or personal data is in it. Three demo accounts let you test each role; their emails and the shared password are in Appendix 1.

! **Your changes are real.** Tests change the demo data (sales, stock, settings). That is fine. If you want a clean start at any time, reset the database, which takes under a minute.

Part A · Set up your computer

You will install three things: **WSL 2** (a small Linux inside Windows), **Docker Desktop** (runs the app's parts, called containers) and **Git** (downloads the code). Then you download StockSense AI and start it with one command.

What you need

ITEM	MINIMUM	BETTER
Windows	Windows 10 version 22H2, 64-bit	Windows 11
Memory (RAM)	8 GB	16 GB
Free disk space	20 GB	30 GB, on an SSD
Processor	4 cores, with virtualization	6 or more cores
Internet	Needed once, to download about 3 GB. After that the app works offline.	

1 Check that virtualization is on

Docker needs the processor's virtualization feature.

1. Press `Ctrl` + `Shift` + `Esc` to open **Task Manager**.
2. Open the **Performance** tab and choose **CPU**.
3. Look for **Virtualization: Enabled** at the bottom right.

> **!** If it says **Disabled**, restart the computer, open the BIOS/UEFI setup (usually `F2`, `F10`, `Del` or `Esc` while it starts), and turn on **Intel VT-x**, **Intel Virtualization Technology**, **AMD-V** or **SVM Mode**. Ask your instructor if you are not sure: do not change other BIOS settings.

2 Install WSL 2

1. Right-click the Start button and choose **Terminal (Admin)** (on Windows 10: **Windows PowerShell (Admin)**). Click **Yes**.
2. Type this and press `Enter`:

```
wsl --install
```

YOU SHOULD SEE

```
Installing: Windows Subsystem for Linux
Installing: Ubuntu
The requested operation is successful. Changes will not be effective until the
system is rebooted.
```

3. **Restart the computer.** After it restarts, an **Ubuntu** window opens and asks for a new UNIX username and password. Choose any short username (lower case) and a password you will remember. This is only for the Linux inside your PC.
4. Open Terminal again (it does not need to be Admin this time) and check:

```
wsl --status
```

YOU SHOULD SEE

```
Default Distribution: Ubuntu
Default Version: 2
```

i If WSL is already installed, `wsl --install` says so. Run `wsl --update` to make sure it is current.

3 Install Docker Desktop

1. Go to docker.com/products/docker-desktop (https://www.docker.com/products/docker-desktop/) and download **Docker Desktop for Windows (AMD64)**, version **4.27 or newer** (if you already have an older one, update it from its whale menu). Use ARM64 only if your laptop has an ARM (Snapdragon) processor.
2. Run the installer. Keep **Use WSL 2 instead of Hyper-V** ticked. When it finishes, restart if it asks you to.
3. Start **Docker Desktop** from the Start menu. Accept the terms. You can skip signing in. Wait until the bottom-left corner says **Engine running** (the whale icon in the taskbar stops moving).
4. Open **Settings** (the gear) → **Resources** → **WSL integration**, turn on **Ubuntu**, and click **Apply & restart**.

Check it from a terminal:

```
docker version
docker compose version
```

YOU SHOULD SEE

```
Client:
 Version:           28.x.x
 ...
```

```
Server: Docker Desktop
 Engine:
  Version:          28.x.x
 ...
Docker Compose version v2.x.x
```

! If `docker version` shows **Client** but then an error such as `error during connect` or `cannot connect to the Docker daemon`, Docker Desktop is not running yet. Start it and wait for **Engine running**.

i **Only 8 GB of RAM?** Docker can use up to half of it by default, which is enough. Close other heavy apps (games, many browser tabs) while testing.

4 Install Git

Inside Ubuntu, Git is usually already there. Open the **Ubuntu** app from the Start menu and check:

```
git --version
```

YOU SHOULD SEE

```
git version 2.x.x
```

If it says `command not found`, install it:

```
sudo apt update && sudo apt install -y git
```

5 Download StockSense AI

In the **Ubuntu** window (not PowerShell), download the code into your Linux home folder. Keeping it inside Linux makes the app much faster than a folder on `C:` or `D:`.

```
cd ~
git clone https://github.com/KxRaxe/stocksense-ai.git
cd stocksense-ai
ls
```

YOU SHOULD SEE

```
CHANGELOG.md  README.md  compose.prod.yaml  compose.yaml  contracts  docker  do
cs  e2e  ml  tests  web
```

> **i** **Prefer a Windows folder?** It also works: run the same commands in PowerShell inside, for example, `C:\Projects`. The first page load is then slower (up to about 30 seconds after each start) because Windows folders are slow for containers. Everything else in this manual is the same.

6 Build and start the app

Still in the `stocksense-ai` folder, run:

```
docker compose up -d --build
```

The first time, this downloads the base images and builds the app, then installs the PHP and JavaScript packages inside the containers. **Allow 10 to 20 minutes** on a typical connection. Later starts take under a minute.

YOU SHOULD SEE

```
[+] Building ...
[+] Running 9/9
 ✓ Container stocksense-db-1          Healthy
 ✓ Container stocksense-redis-1       Healthy
 ✓ Container stocksense-mailpit-1     Healthy
 ✓ Container stocksense-ml-1          Healthy
 ✓ Container stocksense-app-1         Healthy
 ✓ Container stocksense-horizon-1     Started
 ✓ Container stocksense-scheduler-1   Started
 ✓ Container stocksense-vite-1        Started
 ✓ Container stocksense-web-1         Started
```

Check that everything is up:

```
docker compose ps
```

Every service should say **Up**, and those with a health check **(healthy)**. On the very first start `app` installs its packages, which takes a few minutes; until then it says **(health: starting)** and `web`, `vite`, `horizon` and `scheduler` wait for it. Run the command again after a few minutes. What each container does:

CONTAINER	WHAT IT IS
`web`	The web server you open in the browser (nginx), port 8080
`app`	The application itself (Laravel, PHP)
`vite`	Builds the screens (React) while you use them
`horizon`	Background worker: imports, forecasts, emails
`scheduler`	Runs timed jobs: daily advice, digests, weekly forecasts
`ml`	The forecasting service (Python, XGBoost)
`db`	The database (PostgreSQL 17)
`redis`	The queue and cache
`mailpit`	A pretend mailbox that catches every email the app sends

7 Create the database and load the demo shop

```
docker compose exec app php artisan migrate
docker compose exec app php artisan db:seed
```

YOU SHOULD SEE

```
   INFO  Running migrations.
  ...
  2026_10_04_100000_create_settings_table ........ 20.92ms DONE

Loading categories and products...
Loading deliveries...
Importing sales through the import pipeline...
  30,084 sales imported (0 skipped as duplicates, 0 failed).
  Database\Seeders\DemoDataSeeder ............... 37,075 ms DONE
```

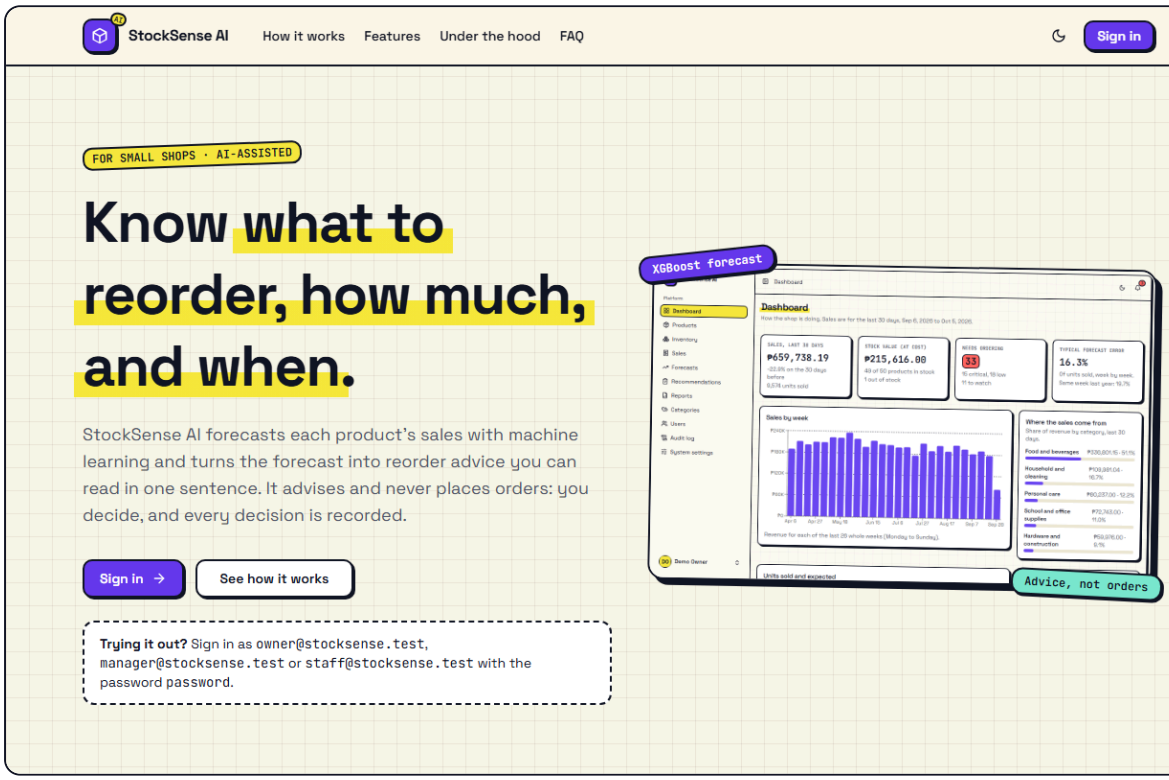
Migrating takes under half a minute; seeding about a minute: it loads the 50 products and about 30,000 sales through the app's own importer.

8 Open the app

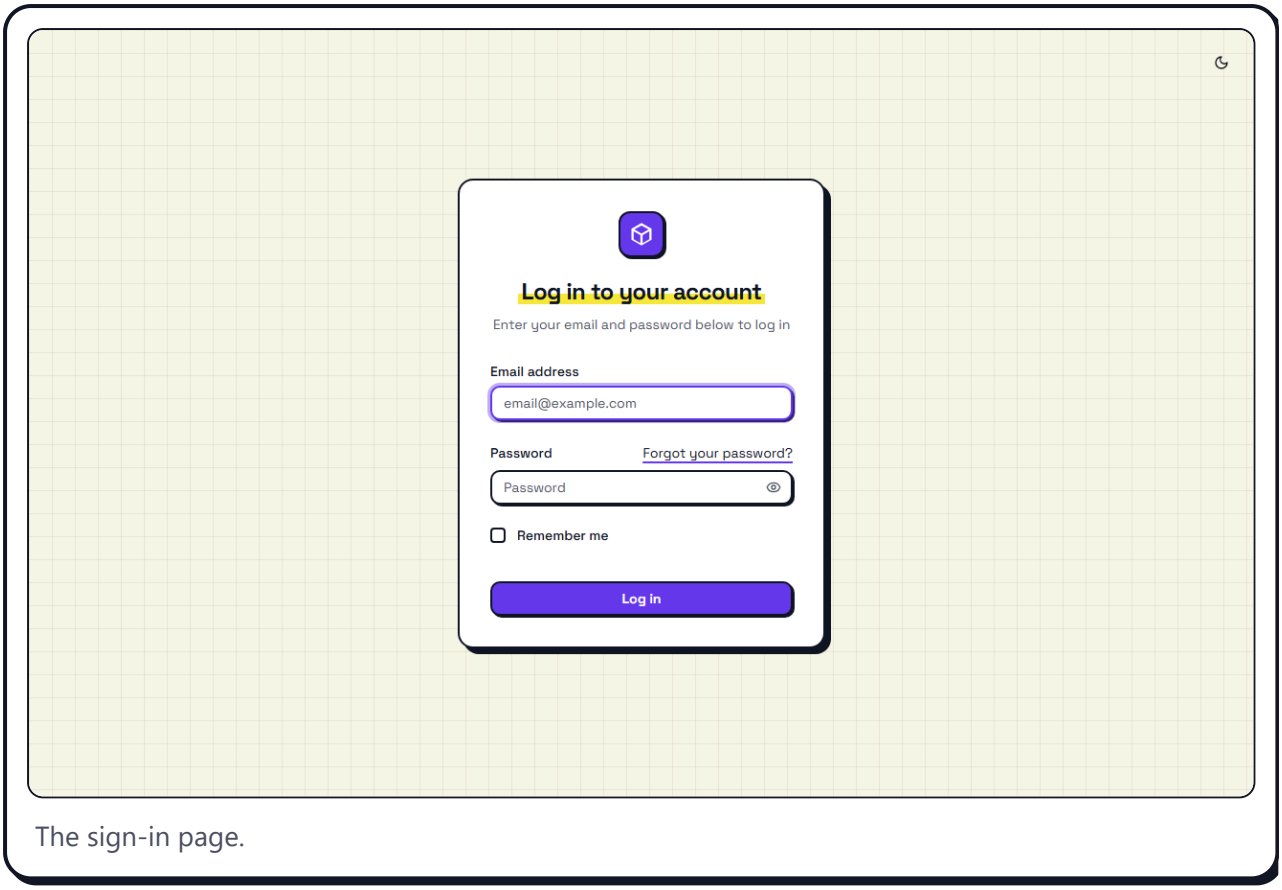
WHAT	ADDRESS
StockSense AI	http://localhost:8080
Mailpit (the pretend mailbox)	http://localhost:8026
Forecasting service API docs	http://localhost:8001/docs

The **first** page after a start can take up to a minute while the screens are built, and stays blank until then. That is normal. Later pages are quick.

You land on the **front page**, which explains what StockSense AI does. Click **Sign in** (top right) to reach the sign-in page.



The front page. The moon/sun button next to Sign in switches between light and dark mode.



The sign-in page.

9 Stop and start it again later

```
# stop (your data is kept)
docker compose stop

# start again
docker compose start
```

Docker Desktop must be running first. You do not need to build or seed again.

✓ **Ready.** If the front page appears, set-up is done. Go to the smoke test.

Troubleshooting

WHAT YOU SEE	WHAT TO DO
`Bind for 0.0.0.0:8080 failed: port is already allocated`	Another program uses that port (often another web server or an old copy of this app). Stop it, or find it with `netstat -ano | findstr :8080` in PowerShell. The app also uses ports 5173, 5433, 8001 and 8026.
The page stays blank or grey for over a minute	Run `docker compose logs vite --tail 30`. If it is still installing, wait. If it shows an error, run `docker compose restart vite`.
`502 Bad Gateway`	The app container is still starting. Wait a minute and reload. If it stays, run `docker compose ps` and `docker compose logs app --tail 50`.
Docker Desktop says **Engine stopped**, or commands say `cannot connect`	Restart Docker Desktop from its whale menu. If it keeps stopping, your PC may be short of memory: close other apps, or lower Docker's limit (below).

Everything is very slow, the fan is loud	Docker may be using too much memory. Create the file `C:\Users\<you>\.wslconfig` with the two lines below, then run `wsl --shutdown` in PowerShell and start Docker Desktop again.
	`[wsl2]`
	`memory=6GB`

`exec /usr/local/bin/entrypoint.sh: no such file or directory`	The files were downloaded with Windows line endings. Delete the folder and clone again from the Ubuntu window (step 5), or run `git config --global core.autocrlf false` first and clone again.
Login says the email or password is wrong for the demo accounts	The demo data was not loaded. Run step 7 again (`db:seed`).
A forecast stays on “running” for more than 5 minutes	Run `docker compose logs horizon --tail 50` and `docker compose logs ml --tail 50`, then `docker compose restart horizon`, and try again.
No emails arrive in Mailpit	Emails are sent by the background worker. Check `docker compose ps horizon` says **Up**; if not, `docker compose up -d horizon`.

Reset to a clean demo shop

This deletes everything you changed and loads the demo shop again (under a minute):

```
docker compose exec app php artisan migrate:fresh --seed
```

To remove the app completely, including its data: `docker compose down -v`, then delete the `stocksense-ai` folder.

Part B · Test cases

143 test cases in 19 groups. Sign in as the role shown on each case (accounts). Unless a case says otherwise, start from the dashboard.

B1. Smoke test (5 minutes)

A quick check that the whole system is alive before the detailed tests. If any of these fail, fix the set-up first (see Troubleshooting).

SMK-01 **All containers are running** — Signed out

Before you start: Part A is done.

1. In the `stocksense-ai` folder run `docker compose ps`.

EXPECTED RESULT
Nine services are listed. All say **Up**; `app`, `db`, `redis`, `ml`, `mailpit`, `web` and `vite` also say **(healthy)**.

☐ Pass ☐ Fail ☐ Blocked Remarks

SMK-02 **The app answers** — Signed out

1. Open `http://localhost:8080`.

EXPECTED RESULT
Within about a minute the front page appears: the headline **Know what to reorder, how much, and when.**, a **Sign in** button and a picture of the dashboard, on a cream grid background.

☐ Pass ☐ Fail ☐ Blocked Remarks

SMK-03 The forecasting service answers

Signed out

1. Open `http://localhost:8001/health`.

EXPECTED RESULT

The page shows a short JSON text that contains `"status":"ok"`.

☐ Pass ☐ Fail ☐ Blocked Remarks

SMK-04 The pretend mailbox answers

Signed out

1. Open `http://localhost:8026`.

EXPECTED RESULT

The Mailpit inbox opens. It may be empty.

☐ Pass ☐ Fail ☐ Blocked Remarks

SMK-05 A demo account can sign in

Signed out

Before you start: The demo shop is loaded (Part A, step 7).

1. On the front page click **Sign in**.
2. Enter `owner@stocksense.test` and `password`, then click **Log in**.

EXPECTED RESULT

The **Dashboard** opens with four figure cards (sales, stock value, needs ordering, forecast error) and charts. The menu on the left shows Dashboard, Products, Inventory, Sales, Forecasts, Recommendations, Reports, Categories, Users, Audit log and System settings.

☐ Pass ☐ Fail ☐ Blocked Remarks

SMK-06 Make the first forecast

Owner

Before you start: SMK-05 is done. A new copy has no forecast yet, so **Needs ordering** shows 0 and there is no advice. Many later tests need one.

1. Open **Forecasts** and click **Run weekly forecast**.
2. Wait until the summary appears (usually under a minute, up to a few minutes on a slow computer).
3. Open the **Dashboard** again.

EXPECTED RESULT

The forecasts page shows a summary and a table of 50 products. On the dashboard, **Needs ordering** now shows a coloured number and **Typical forecast error** a percentage. **Recommendations** lists cards.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

B2. The front page

What a visitor sees at `http://localhost:8080` before signing in: what the app does, for shop owners and for the curious.

LND-01 It explains the app

Signed out

Before you start: You are signed out.

1. Open `http://localhost:8080`.
2. Scroll through the whole page.

> EXPECTED RESULT
> In order: the headline and a dashboard picture; **Stock is a guessing game** (three problems); **How it works** (four numbered steps and an example of the advice); **Everything a small shop needs** (eight features); **Under the hood** (the forecast, privacy and security, the technologies); **One shop, three roles**; **Good to know** (questions); and a violet **Ready when you are** box. It says clearly that the app never places orders.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

LND-02 Every Sign in button leads to the sign-in page

Signed out

Before you start: You are signed out.

1. Click **Sign in** in the top bar. Go back. Click **Sign in** next to the headline. Go back. Click **Sign in** in the violet box at the bottom.

> EXPECTED RESULT
> Each one opens the sign-in page.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

LND-03 The top links and the questions

Signed out

Before you start: You are signed out, on a computer screen (the top links are hidden on a phone).

1. Click **How it works**, **Features**, **Under the hood** and **FAQ** in the top bar.
2. Under **Good to know**, click each question.

> EXPECTED RESULT
> Each link scrolls to its section. Each question opens to show its answer, and closes again when clicked.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

LND-04 Demo accounts are offered only on a test copy

Signed out

Before you start: You are signed out.

1. Look under the two buttons next to the headline.

> EXPECTED RESULT
> A dashed box lists the three demo accounts and the password. (On a real, production installation this box does not appear, because the demo accounts do not exist there.)

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

LND-05 Signed in, you skip the front page

Any role

Before you start: You are signed in.

1. Open `http://localhost:8080`.

> EXPECTED RESULT
> You go straight to the **Dashboard**. After you sign out, you return to the front page.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

B3. Signing in and your account

Sign-in, sign-out, password reset by email, profile and security settings. The demo password is `password` for all three accounts.

AUTH-01 Sign in with the right password

Signed out

Before you start: You are signed out.

1. Open `http://localhost:8080/login`.
2. Enter `manager@stocksense.test` / `password` and click **Log in**.

EXPECTED RESULT

The Dashboard opens. Your name, **Demo Manager**, is at the bottom of the left menu.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks _____

AUTH-02 A wrong password is refused, without saying which part was wrong

Signed out

Before you start: You are signed out.

1. Enter `manager@stocksense.test` and the password `wrong-password`.
2. Click **Log in**.

EXPECTED RESULT

You stay on the sign-in page. A red message under the email says the details do not match our records. It does not say whether the email exists.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks _____

AUTH-03 An unknown email is refused the same way

Signed out

Before you start: You are signed out.

1. Enter `nobody@stocksense.test` / `password` and click **Log in**.

EXPECTED RESULT

The same message as in AUTH-02. Nothing tells you whether an account exists.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks _____

AUTH-04 Too many wrong tries pause sign-in

Signed out

Before you start: You are signed out.

1. Try to sign in as `staff@stocksense.test` with a wrong password **six times** within one minute.

EXPECTED RESULT

From the sixth try a message says **Too many login attempts. Please try again in … seconds.** Even the right password is refused until the wait is over (about a minute). Then sign-in works again.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

AUTH-05 Show and hide the password

Signed out

Before you start: On the sign-in page.

1. Type something in the Password field.
2. Click the eye button at the right end of the field. Click it again.

EXPECTED RESULT

The first click shows the password as plain text; the second hides it again.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

AUTH-06 Sign out

Any role

Before you start: You are signed in.

1. Click your name at the bottom of the left menu.
2. Click **Log out**.
3. Press the browser's Back button.

EXPECTED RESULT

You are taken to the front page. Going back does not show the app; any page you open asks you to sign in again.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

AUTH-07 Pages need a signed-in person

Signed out

Before you start: You are signed out.

1. Type `http://localhost:8080/products` in the address bar.

> EXPECTED RESULT
> You are sent to the sign-in page instead of the product list.

☐ Pass ☐ Fail ☐ Blocked Remarks

AUTH-08 Forgot password: the email is sent

Signed out

Before you start: You are signed out.

1. On the sign-in page click **Forgot your password?**.
2. Enter `staff@stocksense.test` and send the request.
3. Open Mailpit (`http://localhost:8026`).

> EXPECTED RESULT
> A green message says **We have emailed your password reset link.** In Mailpit a new password-reset email is addressed to the staff account.

☐ Pass ☐ Fail ☐ Blocked Remarks

AUTH-09 The reset email contains nothing personal

Signed out

Before you start: AUTH-08 is done.

1. Open the email in Mailpit. Read the subject and the whole text. Look at the link (hover over the button).

> EXPECTED RESULT
> The email starts with a plain **Hello**, not a name. Nowhere does it show a person's name or an email address, not even inside the link. It is drawn in the StockSense style: a violet name tag and a white card with a thick dark edge.

☐ Pass ☐ Fail ☐ Blocked Remarks

AUTH-10 Choose a new password from the link

Signed out

Before you start: AUTH-08 is done.

1. Click the button in the email.
2. Type `staff@stocksense.test` as the email (the link does not carry it), and a new password twice, for example `NewPass123`.
3. Submit, then sign in with the new password.

> EXPECTED RESULT
> You are sent to the sign-in page with a message that the password was reset. The new password works; the old one (`password`) no longer does. *Reset the database afterwards, or sign in as staff and set it back under Settings → Security, so other tests still work.*

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

AUTH-11 A reset link works only once

Signed out

Before you start: AUTH-10 is done.

1. Open the same link from the email again and try to set another password.

> EXPECTED RESULT
> The app refuses: the link is invalid or expired.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

AUTH-12 Change your name

Any role

Before you start: Signed in as the Manager.

1. Click your name (bottom left) → **Settings**.
2. On **Profile**, change the name to `Manager Test` and click **Save**.

> EXPECTED RESULT
> A message says **Profile updated.** The new name appears at the bottom of the menu. Change it back to `Demo Manager`.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

AUTH-13 Change your password

Any role

Before you start: Signed in as the Manager.

1. Settings → **Security** (enter your password again if asked).
2. Fill in the current password, then a new password twice, and save.
3. Sign out and sign in with the new password.

EXPECTED RESULT

A message says **Password updated.** The new password works. Set it back to `password` the same way.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

AUTH-14 Wrong current password is refused

Any role

Before you start: Settings → Security.

1. Enter a wrong current password and any new password, and save.

EXPECTED RESULT

Nothing changes. A red message under the current-password field says it is incorrect.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

AUTH-15 Turn on two-factor authentication (optional)

Owner

Before you start: You have an authenticator app on your phone (Google Authenticator, Microsoft Authenticator or similar).

1. Settings → **Security** → turn on two-factor authentication.
2. Scan the QR code with the app and type the 6-digit code it shows.
3. Note the recovery codes. Sign out and sign in again.

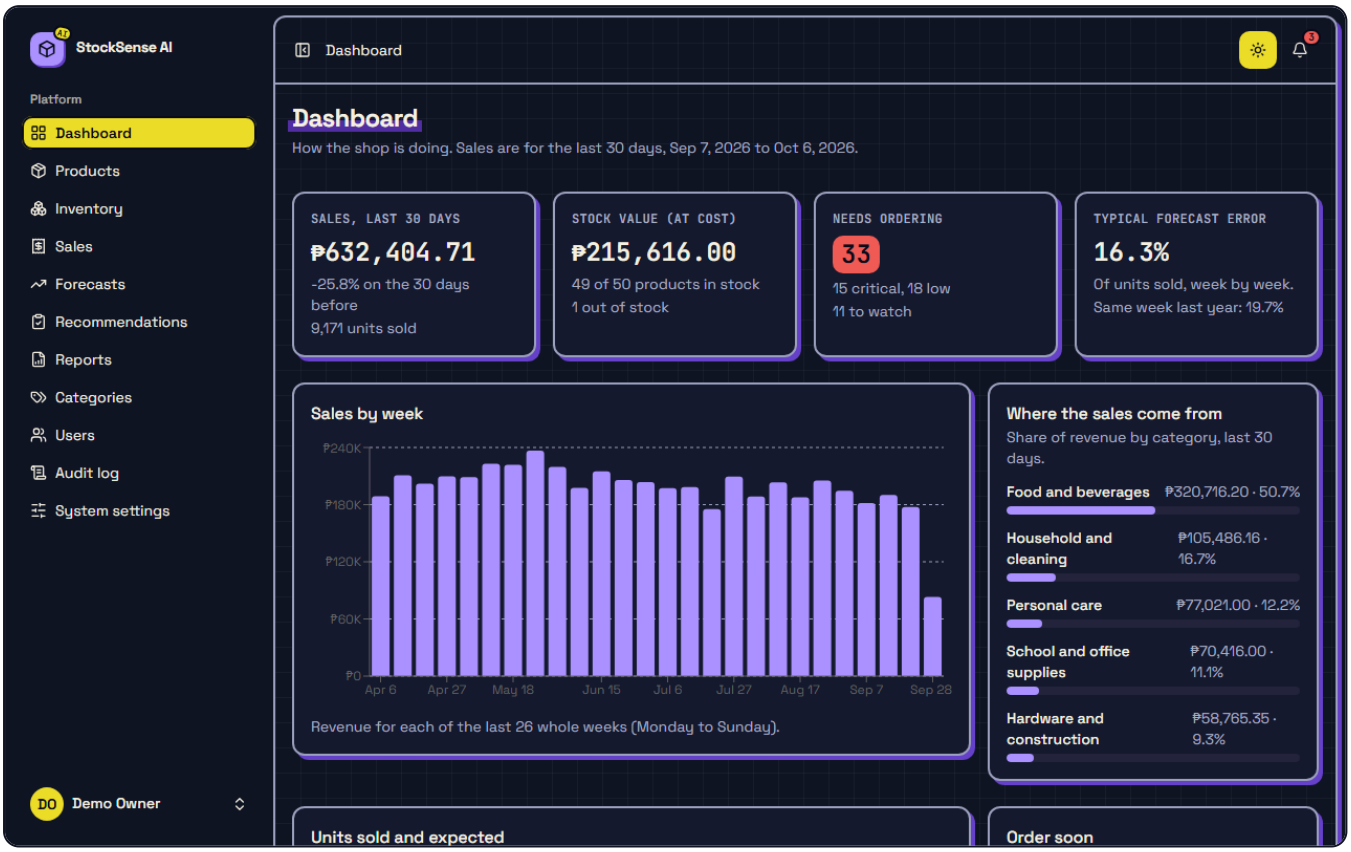
EXPECTED RESULT

After the password the app asks for a 6-digit code. The code from your phone lets you in. Turn two-factor authentication off again at the end, so others can test with this account.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

B4. Light and dark mode

The app has a light theme (cream paper, ink outlines) and a dark theme (deep ink, violet shadows). You can switch with the button at the top right of every page, or choose in Settings → Appearance.



The dashboard in dark mode. The sun/moon button at the top right switches modes.

THM-01 **Switch from the top bar** — Any role

Before you start: Signed in, in light mode.

1. Click the moon button at the top right, next to the bell.

> EXPECTED RESULT
> The whole app turns dark at once: dark background, light text, violet shadows. The button now shows a sun, and its tooltip says **Switch to light mode**.

☐ Pass ☐ Fail ☐ Blocked Remarks _____

THM-02 The choice is remembered

Any role

Before you start: THM-01 is done (dark mode).

1. Open another page, then reload with F5.
2. Sign out and sign in again.

> EXPECTED RESULT
> Every page stays dark, without a flash of light while loading. The sign-in page is dark too.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

THM-03 Switch on the front page and the sign-in page

Signed out

Before you start: Signed out.

1. Click the moon/sun button next to **Sign in** on the front page. Then open the sign-in page and click the button in its top-right corner.

> EXPECTED RESULT
> Both pages switch between light and dark, and the choice carries over from one to the other.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

THM-04 Follow the computer's setting

Any role

Before you start: Signed in.

1. Settings → **Appearance** → choose **System**.
2. In Windows Settings → Personalisation → Colours, switch between Light and Dark.

> EXPECTED RESULT
> The app follows Windows within a moment, without reloading. The chosen tab (Light, Dark or System) is shown in yellow.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

THM-05 **Everything is readable in both modes** Any role

1. In each mode, look at the dashboard, a product, the recommendations and a report.

> EXPECTED RESULT
> All text is easy to read on its background. Badges (Critical, Low, Watch, OK, Overstock, Low stock) are bright with dark text in both modes. Nothing is invisible or hard to read.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

B5. Roles and permissions

There are three roles. Each sees only the menu items it may use, and the server blocks the rest even if someone types the address. Use this table, then check it with the tests below.

WHAT	OWNER	MANAGER	INVENTORY STAFF
Dashboard	✓	✓	✓ **(no forecasts)**
Products and categories	**change**	**change**	look only
Record deliveries and stock counts	✓	✓	✓
Enter and import sales	✓	✓	✓
Run forecasts, see accuracy	✓	✓	–
See recommendations	✓	✓	✓
Decide on recommendations	✓	✓	–
Reports	**all four**	**all four**	stock report only
Users, system settings, audit log	✓	–	–

ROL-01 **The Owner's menu** Owner

Before you start: Signed in as the Owner.

1. Look at the left menu.

> EXPECTED RESULT
> Dashboard, Products, Inventory, Sales, Forecasts, Recommendations, Reports, Categories, Users, Audit log, System settings.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks _____

ROL-02 The Manager's menu

Manager

Before you start: Signed in as the Manager.

1. Look at the left menu.

> EXPECTED RESULT
> The same as the Owner but **without** Users, Audit log and System settings.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

ROL-03 Inventory staff's menu

Inventory staff

Before you start: Signed in as staff.

1. Look at the left menu.

> EXPECTED RESULT
> Dashboard, Products, Inventory, Sales, Recommendations, Reports, Categories. **No** Forecasts, Users, Audit log or System settings.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

ROL-04 Staff cannot open blocked pages by address

Inventory staff

Before you start: Signed in as staff.

1. Type each address in turn: `/forecasts`, `/users`, `/system-settings`, `/audit-log`, `/products/create`, `/reports/sales` (after `http://localhost:8080`).

> EXPECTED RESULT
> Each shows a **403** “forbidden” page. Nothing from those pages is shown.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

ROL-05 The Manager cannot open Owner pages

Manager

Before you start: Signed in as the Manager.

1. Type `/users`, `/system-settings` and `/audit-log`.

> EXPECTED RESULT
> Each shows a 403 page.

☐ Pass ☐ Fail ☐ Blocked Remarks

ROL-06 Staff can look at products but not change them

Inventory staff

Before you start: Signed in as staff.

1. Open **Products**, then any product.

> EXPECTED RESULT
> The list and the product page open. There are no **New product**, **Edit** or **Archive** buttons. **Record restock** and **Count stock** are there.

☐ Pass ☐ Fail ☐ Blocked Remarks

ROL-07 Staff can see advice but not decide

Inventory staff

Before you start: Signed in as staff. A forecast exists.

1. Open **Recommendations**.

> EXPECTED RESULT
> The cards show the advice and reasoning, but there are no **Accept**, **Change quantity**, **Dismiss** or **Recalculate** buttons.

☐ Pass ☐ Fail ☐ Blocked Remarks

ROL-08 Staff sees only the stock report

Inventory staff

Before you start: Signed in as staff.

1. Open **Reports**.

> EXPECTED RESULT
> Only **Inventory status** is offered. Sales, forecast accuracy and replenishment history are not.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks _____

ROL-09 Each role's dashboard

Any role

1. Sign in as each of the three accounts in turn and open the dashboard.

> EXPECTED RESULT
> Owner and Manager see sales, stock value, needs ordering, forecast error and the charts. Staff sees the stock figures and urgent products, without forecast numbers.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks _____

B6. Users

Only the Owner manages people. Nobody is ever sent a password: a new person gets an email link to choose their own.

USR-01 Add a person — Owner

Before you start: Signed in as the Owner. Mailpit is open in another tab.

1. **Users** → **New user**.
2. Name `Test Staff One`, email `test.staff1@stocksense.test`, role **Inventory staff**.
3. Save.

> EXPECTED RESULT
> A message says **User created. A set-up email has been sent.** The person appears in the list. In Mailpit a set-up email arrived for that address.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks _____

USR-02 The set-up email contains nothing personal — Owner

Before you start: USR-01 is done.

1. Open the set-up email. Read it, including the button's link.

> EXPECTED RESULT
> No name and no email address anywhere in the email or in the link. The greeting is a plain **Hello**.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks _____

USR-03 The new person sets a password and signs in — Signed out

Before you start: USR-01 is done. Use a private (incognito) browser window.

1. Click the link in the set-up email.
2. Type the email `test.staff1@stocksense.test` and a password twice.
3. Sign in with it.

> EXPECTED RESULT
> The person signs in and sees the Inventory staff menu (as in ROL-03).

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks _____

USR-04 The same email cannot be used twice

Owner

Before you start: USR-01 is done.

1. Add another user with the email `test.staff1@stocksense.test`.

> EXPECTED RESULT
> Nothing is saved. A message says the email has already been taken.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

USR-05 Required fields

Owner

1. Open **New user** and save without filling anything in.

> EXPECTED RESULT
> Nothing is saved. Name, email and role each show a red message.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

USR-06 Change someone's role

Owner

Before you start: USR-01 is done.

1. Open the user, change the role to **Manager**, save.
2. In the private window, reload the app.

> EXPECTED RESULT
> A message says **User updated.** The person now sees the Manager menu (Forecasts appears).

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

USR-07 Deactivate someone: they are signed out at once

Owner

Before you start: The test user is signed in, in the private window.

1. Open the user and click **Deactivate**, then confirm.
2. In the private window, click any page.

> EXPECTED RESULT
> A message says **User deactivated.** The person is sent to the sign-in page and cannot sign in again (the same message as a wrong password).

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

USR-08 Activate them again

Owner

Before you start: USR-07 is done.

1. Open the user and click **Activate**.

> EXPECTED RESULT
> A message says **User activated.** The person can sign in again.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

USR-09 Send a new set-up link

Owner

1. Open an active user and click **Send set-up link**.

> EXPECTED RESULT
> A message says **Set-up email sent.** A new email arrives in Mailpit.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

USR-10 **The Owner cannot lock themselves out** Owner

1. Open your own user (Demo Owner).

> EXPECTED RESULT
> There is no Deactivate button for yourself, and the role cannot be changed (the form says **You cannot change your own role.**).

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

B7. Categories

Categories group products and set a **service level**: how often you want to have stock when a customer asks (higher means more safety stock).

CAT-01 Add a category — Manager

1. **Categories** → **New category**.
2. Name `Test pet supplies`, service level `90`.
3. Click **Create category**.

> EXPECTED RESULT
> A message says **Category created.** It appears in the list with 90%.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks _____

CAT-02 Names are unique, whatever the capitals — Manager

Before you start: CAT-01 is done.

1. Add another category named `TEST PET SUPPLIES`.

> EXPECTED RESULT
> Nothing is saved. A red message says the name is already taken.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks _____

CAT-03 The service level must be between 50 and 99.9 — Manager

1. Try service levels `40`, then `100`, then `99.9`.

> EXPECTED RESULT
> 40 and 100 are refused with a red message; 99.9 is accepted.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks _____

CAT-04 Edit a category

Manager

Before you start: CAT-01 is done.

1. Edit **Test pet supplies**, change the service level to 95, click **Save changes**.

> EXPECTED RESULT
> A message says **Category updated.** The list shows 95%.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks _____

CAT-05 A category in use cannot be deleted

Manager

1. Try to delete **Food and beverages**.

> EXPECTED RESULT
> It is not deleted. The message says **This category still has products, including archived ones. Move them to another category first.**

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks _____

CAT-06 Delete an empty category

Manager

Before you start: CAT-01 is done and the category has no products.

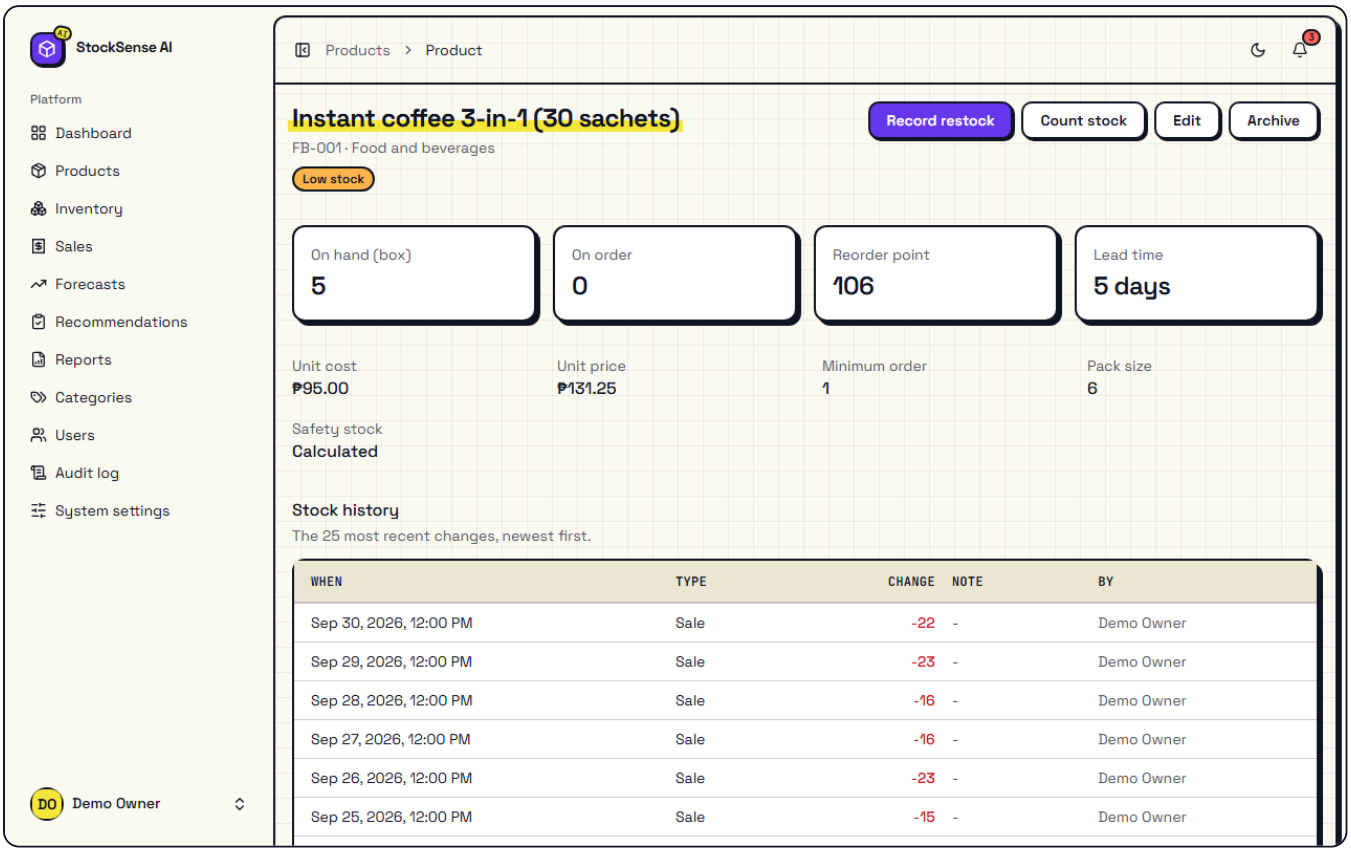
1. Delete **Test pet supplies** and confirm.

> EXPECTED RESULT
> A message says **Category deleted.** It is gone from the list.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks _____

B8. Products

Each product has a SKU (the shop's own code, stored in capitals), cost and price, and reordering details. Products are never deleted, only archived, so their history stays.



A product page: stock figures, the details, the stock ledger, and buttons to record a delivery or a stock count.

PRD-01 **Search and filter the product list** — Any role

1. Open **Products**.
2. Type `soap` in the search box and wait a moment.
3. Clear it and choose the category **Hardware and construction**.

> EXPECTED RESULT
> The list narrows as you type (no button needed) to the soaps. Then only hardware products are shown. The address bar changes, so the filtered list can be bookmarked.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks _____

PRD-02 Add a product

Manager

1. **Products** → **New product**.
2. SKU `ts-001`, name `Test notebook A5`, category **School and office supplies**, unit `pc`, cost `25`, price `39`, lead time `4`, minimum order `1`, pack size `10`, opening stock `50`.
3. Click **Create product**.

> EXPECTED RESULT
> A message says **Product created.** Its page shows SKU **TS-001** (in capitals) and 50 on hand. The stock ledger has one line: opening stock 50.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks _____

PRD-03 SKUs are unique, whatever the capitals

Manager

Before you start: PRD-02 is done.

1. Add another product with SKU `TS-001` (or `ts-001`).

> EXPECTED RESULT
> Nothing is saved; a red message says the SKU is already taken.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks _____

PRD-04 Checks on the numbers

Manager

1. On **New product** try: price `-5`; lead time `400`; pack size `0`; leave the name empty.

> EXPECTED RESULT
> Each is refused with a red message next to the field: price cannot be negative, lead time at most 365 days, pack size at least 1, name is required.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks _____

PRD-05 Edit a product

Manager

Before you start: PRD-02 is done.

1. Open TS-001 → **Edit**. Change the price to `42`, save.

> EXPECTED RESULT
> A message says **Product updated.** The page shows ₱42.00.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

PRD-06 Archive and restore

Manager

Before you start: PRD-02 is done.

1. Open TS-001 → **Archive**, confirm.
2. Find it with the list's status filter set to archived, open it, click **Restore**.

> EXPECTED RESULT
> **Product archived.**: it disappears from the normal list but its history stays. **Product restored.**: it is back.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

PRD-07 Import a product list

Manager

Before you start: Save the file below as `products-test.csv` (Notepad → Save as → All files).

1. **Products** → **Import from a file**. (You can also click **Download an example file** to see the layout.)
2. Choose the file and click **Upload and check**.
3. Read the preview, then click the **Import 2 rows** button.

EXPECTED RESULT

The preview shows **Will be created** 2, with the columns matched automatically. After the import, both products exist with their opening stock.

```
sku,name,category,unit,unit_cost,unit_price,lead_time_days,moq,pack_size,reord
er_point,opening_stock
TS-101,Test glue stick,School and office supplies,pc,12.00,20.00,5,1,12,,36
TS-102,Test sponge,Household and cleaning,pc,9.50,15.00,4,1,10,,40
```

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

PRD-08 A product file with mistakes

Manager

Before you start: Save this as `products-bad.csv`.

1. Import it as in PRD-07.

EXPECTED RESULT

The preview shows **Will be created** 1 and **Have a problem, skipped** 2, and lists the problems row by row (unknown category; a negative price). Only the good row is imported. A downloadable list of the failed rows is offered.

```
sku,name,category,unit,unit_cost,unit_price,lead_time_days,moq,pack_size,reord
er_point,opening_stock
TS-201,Good row,Personal care,pc,10.00,18.00,5,1,1,,5
TS-202,Bad category,Garden tools,pc,10.00,18.00,5,1,1,,5
TS-203,Bad price,Personal care,pc,10.00,-18.00,5,1,1,,5
```

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

PRD-09 Undo a product import

Manager

Before you start: PRD-07 is done.

1. Open the import (Products → Import from a file → the list of past imports) and click **Undo this import**.

> EXPECTED RESULT
> The products the import created are archived. A message says **Import undone.**

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

B9. Stock and inventory

Stock is a ledger: every change is a line with who, when and why, and the stock on hand is the total of those lines.

INV-01 The stock overview

Inventory staff

1. Open **Inventory**.

> EXPECTED RESULT
> Every product with what is on hand, on order and its reorder point. Products at or below the reorder point have an orange **Low stock** badge; products with nothing left have a red **Out of stock** badge.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks _____

INV-02 Record a delivery (restock)

Inventory staff

1. Open any product → **Record restock**.
2. Quantity `24`, today's date, note `Test delivery`. Click **Save restock**.

> EXPECTED RESULT
> A message says **Restock recorded.** On hand goes up by 24. The ledger has a new line with +24, your name and the note.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks _____

INV-03 A delivery cannot be in the future

Inventory staff

1. Record restock with tomorrow's date.

> EXPECTED RESULT
> Refused: the date cannot be in the future.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks _____

INV-04 Count stock

Inventory staff

Before you start: Note the product's current on-hand number.

1. Same product → **Count stock**.
2. Enter a number 3 lower than on hand, reason `Test count: damaged`. Click **Save count**.

> EXPECTED RESULT
> A message says **Stock adjusted.** On hand now equals what you counted. The ledger shows the difference (−3) with your reason.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks _____

INV-05 A count needs a reason

Inventory staff

1. Count stock without typing a reason.

> EXPECTED RESULT
> Refused: a reason is required.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks _____

INV-06 Low-stock flag follows the reorder point

Manager

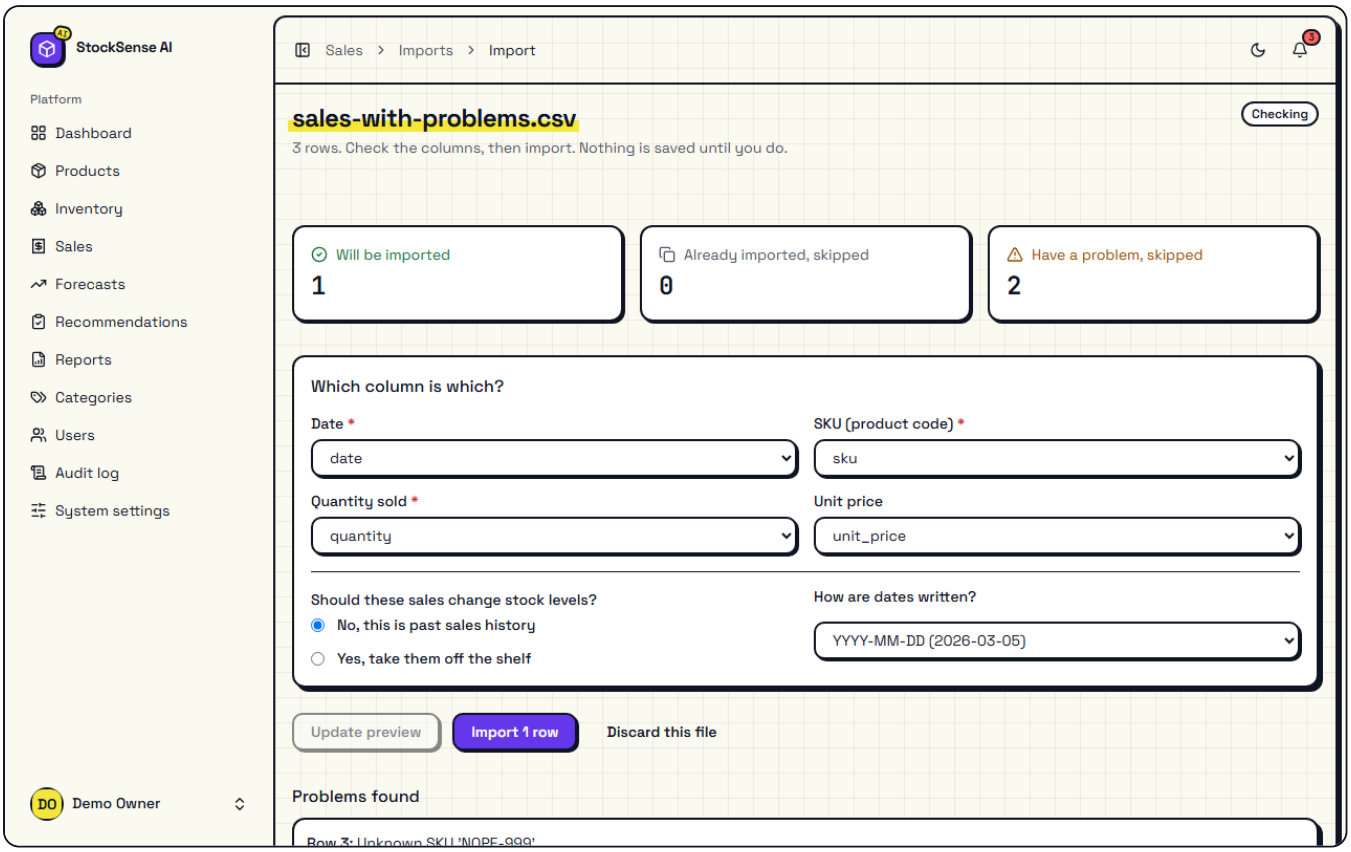
1. Pick a product with a reorder point. Count its stock to a number **below** the reorder point.
2. Open Inventory.

> EXPECTED RESULT
> The product now shows the **Low stock** badge.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks _____

B10. Sales and imports

Sales are what the forecasts learn from. You can type them in or import a file. **Use yesterday's date** in the files below (a sale cannot be in the future). Write dates as `YYYY-MM-DD`, for example `2026-10-04`.



An import preview: what will be imported, what will be skipped as a repeat, and what is wrong.

SAL-01 Enter sales by hand

Inventory staff

Before you start: Note the on-hand stock of **FB-006** (Instant noodles).

1. **Sales → Record sales**.
2. Date: yesterday. Line 1: product FB-006, quantity 2. Click **Add a line**; line 2: FB-007, quantity 1.
3. Save.

> EXPECTED RESULT
> A message says **Sale recorded.** The sales list shows both lines. FB-006's stock went down by 2.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks _____

SAL-02 Warning when selling more than is in stock

Inventory staff

1. In **Record sales**, pick a product and type a quantity larger than its stock.

> EXPECTED RESULT
> A warning under the line says it is more than is in stock. You can still save (the shelf count may be wrong).

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

SAL-03 Checks on a sale

Inventory staff

1. Try a quantity of `0`, then a date in the future.

> EXPECTED RESULT
> Both are refused with red messages.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

SAL-04 Delete a sale

Manager

Before you start: SAL-01 is done.

1. In the sales list, delete one of your test sales and confirm.

> EXPECTED RESULT
> A message says **Sale deleted and its stock put back.** The product's stock goes back up.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

SAL-05 Search and filter sales

Any role

1. In **Sales**, search `FB-006`; then set From and To to last week; then set the source to imported.

> EXPECTED RESULT
> The list follows each filter straight away.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

SAL-06 Import a sales file: preview first

Manager

Before you start: Save the file below as `sales-test.csv`, with yesterday's date.

1. **Sales → Import from a file**.
2. Choose the file. For **Should these sales change stock levels?** choose **Yes, take them off the shelf**. Click **Upload and check**.
3. In the preview, check that **How are dates written?** says **YYYY-MM-DD**.

EXPECTED RESULT

A preview: **Will be imported** 3, **Already imported, skipped** 0, **Have a problem, skipped** 0. Under **Which column is which?** the columns are matched to Date, SKU, Quantity sold and Unit price. **Nothing is saved yet.**

```
date,sku,quantity,unit_price
2026-10-04,FB-006,4,69.30
2026-10-04,PC-003,2,68.00
2026-10-04,HW-004,1,719.25
```

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

SAL-07 Start the import

Manager

Before you start: SAL-06 is done.

1. Click **Import 3 rows**.
2. Wait for the result.

EXPECTED RESULT

The page shows progress, then the result: **Rows in the file** 3, **Imported** 3, **Already there, skipped** 0, **Failed** 0. The rows appear in **Sales** with the source **Imported from a file**, and the three products' stock went down.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

SAL-08 The same file twice: repeats are skipped

Manager

Before you start: SAL-07 is done.

1. Import exactly the same file again.

> EXPECTED RESULT
> The preview shows **Already imported, skipped** 3 and **Will be imported** 0. Nothing is imported twice.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

SAL-09 A file with problems

Manager

Before you start: Save this as `sales-bad.csv` (yesterday's date on line 2).

1. Import it.
2. After the import, click **Download the error report**.

> EXPECTED RESULT
> **Have a problem, skipped** shows 4, and **Problems found** lists each one with its row: unknown SKU `NOPE-999`, a date that does not exist, a quantity that is not a number, a negative quantity. Only the good row is imported. The error report is a CSV of the failed rows with the reason.

```
date,sku,quantity,unit_price
2026-10-04,FB-006,5,69.30
2026-10-04,NOPE-999,1,10.00
2026-13-45,FB-007,2,288.75
2026-10-04,FB-008,many,35.70
2026-10-04,FB-010,-3,162.75
```

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

SAL-10 Undo an import

Manager

Before you start: SAL-07 is done.

1. Open that import and click **Undo this import**, then confirm.

> EXPECTED RESULT
> A message says **Import undone.** Its sales are gone from the list, and the stock it took off is put back.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

SAL-11 Choose columns yourself

Manager

Before you start: A file whose headers are unusual, for example `Fecha,Item,Qty`.

1. Upload it. Under **Which column is which?** set Date → Fecha, SKU → Item, Quantity sold → Qty. Click **Update preview**.

> EXPECTED RESULT
> Until all required columns are chosen, the preview says which are missing and cannot be started. After choosing them, the rows are checked again.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

SAL-12 Wrong kind of file

Manager

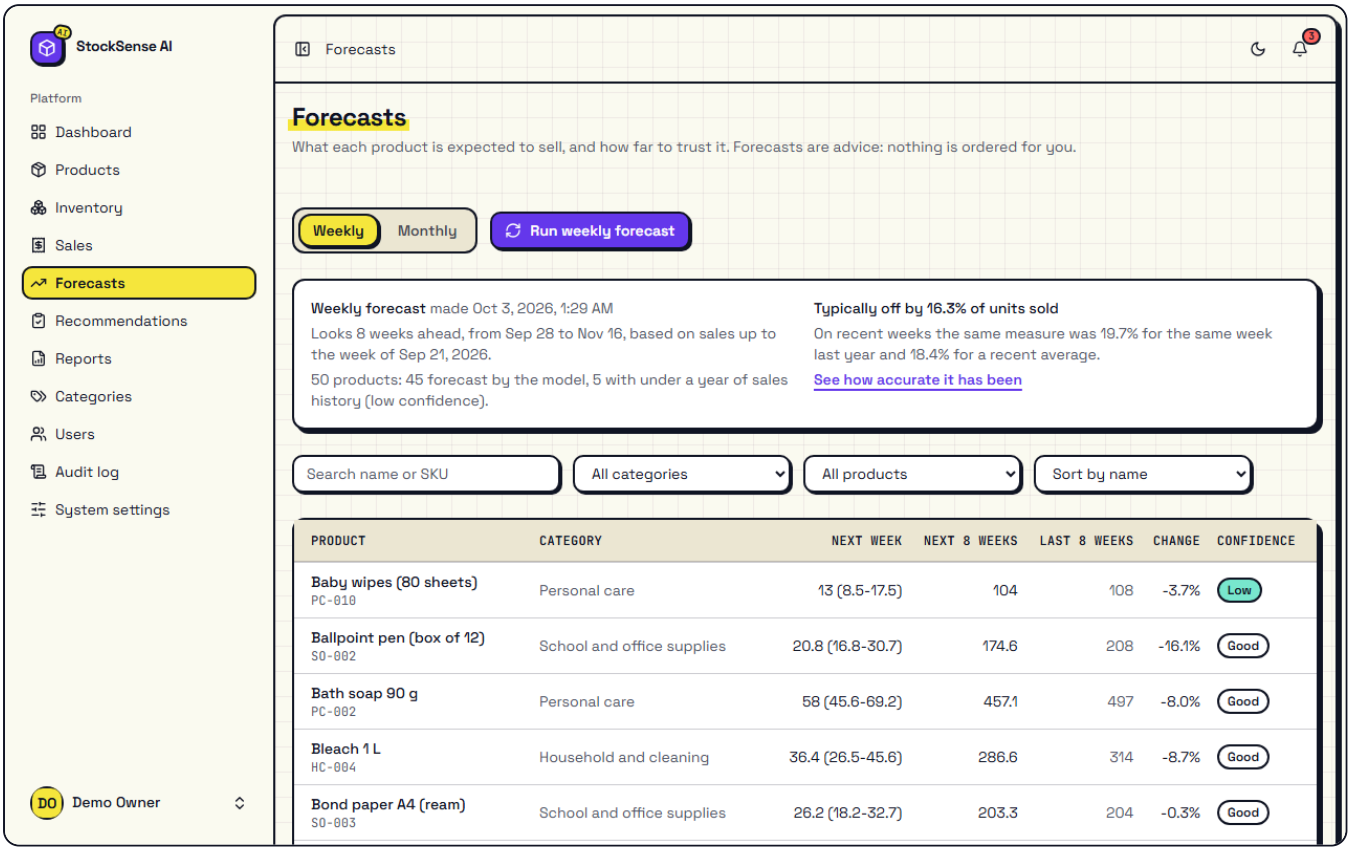
1. Try to upload a `.pdf` or `.docx` file.

> EXPECTED RESULT
> Refused: only CSV, TXT or Excel (.xlsx) files up to 10 MB.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

B11. Forecasts

The forecasting model (XGBoost) predicts each product's sales, week by week or month by month, with a likely range. Products with under a year of history get a simpler average and a **Low confidence** label. A run trains a real model: from under a minute to a few minutes, depending on the computer.



The forecasts page after a weekly run: how far ahead it looks, how accurate it has been, and every product's forecast.

FOR-01 Run a weekly forecast

Manager

1. Open **Forecasts**. Make sure **Weekly** is selected.
2. Click **Run weekly forecast**.
3. Wait; you may open other pages meanwhile and come back.

> EXPECTED RESULT
>
> A message says **Forecast started. It usually takes a minute or two.** A progress box appears and updates by itself. When done, a summary shows how far ahead it looks (8 weeks), how many products were forecast (50, of which some are low confidence) and **Typically off by ...% of units sold**. A table lists every product.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

FOR-02 Only one run at a time

Manager

Before you start: Start a weekly run and, while it runs, click the button again (or open the page in a second tab and click there).

1. Try to start a second weekly run.

> EXPECTED RESULT
> The second is refused: **A weekly forecast is already running.**

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

FOR-03 Run a monthly forecast

Manager

1. Click **Monthly**, then **Run monthly forecast**. Wait for it.

> EXPECTED RESULT
> The same as FOR-01, looking 3 months ahead. The address bar now ends in `?granularity=month`.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

FOR-04 One product's forecast and chart

Manager

Before you start: A weekly forecast exists.

1. In the forecasts table, click a product's name.

> EXPECTED RESULT
> A chart shows recent weekly sales followed by the forecast line and a shaded band (the likely range). Below, a table gives each future week with its forecast and range.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

FOR-05 Low-confidence products are marked

Manager

Before you start: A weekly forecast exists.

1. In the forecasts filters, choose **Low confidence only**.

> EXPECTED RESULT
> About 5 newer products are labelled **Low confidence**. Opening one says it has less than a year of history, so a plain recent average is used.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

FOR-06 Accuracy against simple guesses

Manager

Before you start: A weekly forecast exists.

1. Click **See how accurate it has been**.

> EXPECTED RESULT
> The page compares the model with two simple guesses, **the same week last year** and **a recent average**, using MAE, RMSE, MAPE and WAPE (lower is better). The model's WAPE should be lower than both. Below: accuracy per category and what the model pays most attention to.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

FOR-07 Search and filter forecasts

Manager

1. Search `rice`; then choose a category; then change the sort order.

> EXPECTED RESULT
> The table follows each choice straight away.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

FOR-08 Notified when a forecast finishes

Manager

Before you start: FOR-01 is done.

1. Look at the bell at the top right. Open Mailpit.

> EXPECTED RESULT
> The bell has a new notification that the weekly forecast is ready, with its accuracy. Mailpit has the same as an email, sent to you and to the Owner, with no names in it.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

FOR-09 The forecasting service is protected

Signed out

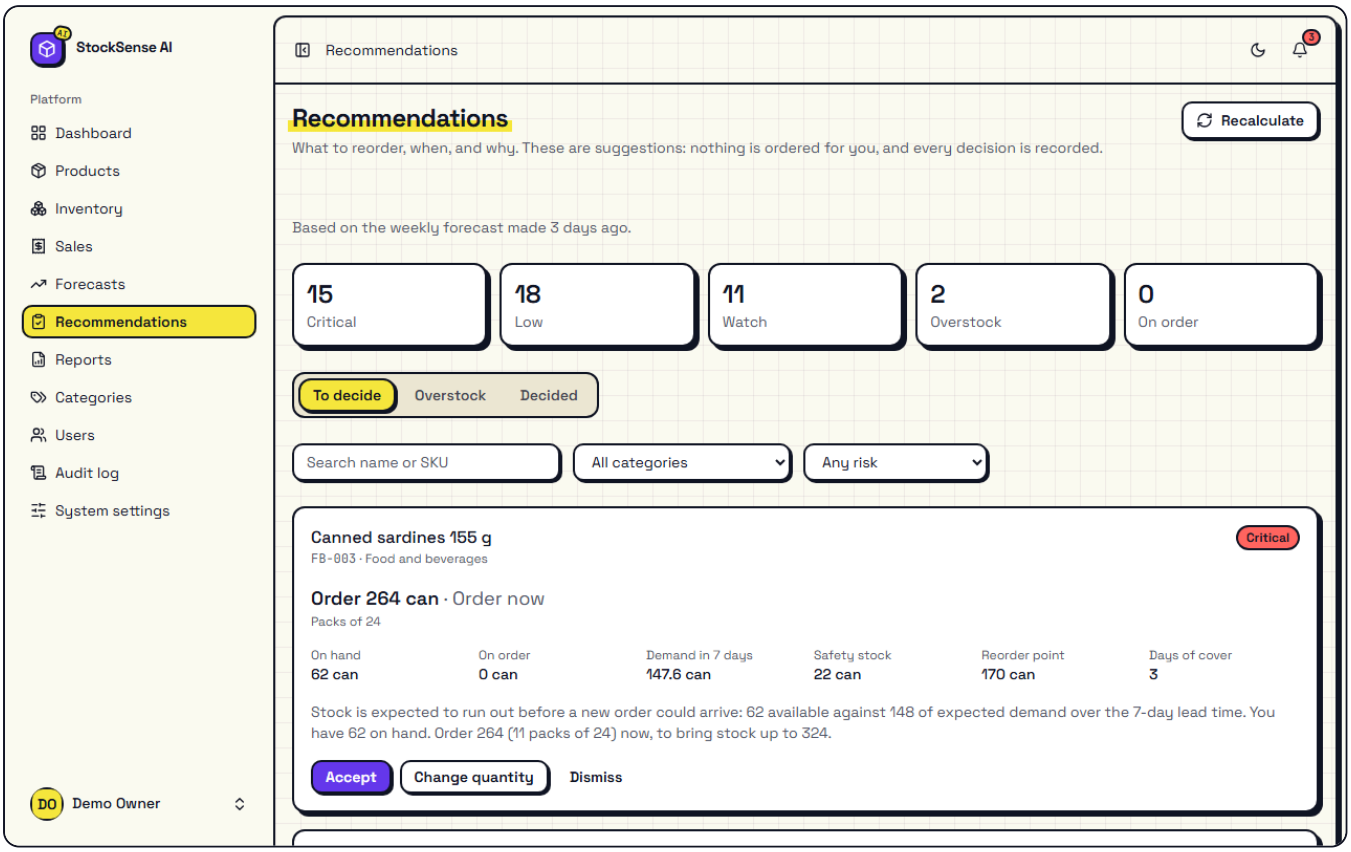
1. Open `http://localhost:8001/docs`. Expand **GET /models**, click **Try it out**, then **Execute** (leave the token empty).

> EXPECTED RESULT
> The response code is **401**: the service refuses anyone without the secret token. Only the app knows it.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

B12. Recommendations

The advice: what to order, how much and by when, with a risk level and a sentence explaining why. The app never orders anything; it only records what you decide.



Recommendations: counts by risk level, the To decide / Overstock / Decided tabs, filters, and one card per product.

REC-01 Advice is ready after a forecast

Manager

Before you start: FOR-01 is done (advice is worked out right after each forecast).

1. Open **Recommendations**.

> EXPECTED RESULT
> Counts at the top: Critical, Low, Watch, Overstock and On order. The page says which forecast the advice is based on and how old it is. Cards are listed most urgent first.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

REC-02 A card explains itself

Manager

1. Read the first card.

EXPECTED RESULT

It shows the product, a risk badge, **Order N … · Order now** (or **Order by** a date), packs, on hand, on order, demand during the lead time, safety stock, reorder point, days of cover, and a sentence such as "Stock is expected to run out before a new order could arrive…".

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

REC-03 Risk badges have the right colours

Any role

1. Look at the badges on several cards and on the Overstock tab.

EXPECTED RESULT

Critical red, **Low** orange, **Watch** blue, **OK** green, **Overstock** purple; each with dark text and readable in light and dark mode.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

REC-04 Accept a recommendation

Manager

1. On a card click **Accept**. Type a note such as `Ordered by phone`. Confirm.

EXPECTED RESULT

A message says **Accepted. The quantity now counts as on order.** The card leaves the To decide tab; the On order count goes up by 1. On the product's page, **On order** shows that quantity.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

REC-05 Accept with a different quantity

Manager

1. On another card click **Change quantity**, type a different number, confirm.

EXPECTED RESULT

A message says **Accepted with your quantity. It now counts as on order.**

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

REC-06 Dismiss a recommendation

Manager

1. On another card click **Dismiss** and confirm.

EXPECTED RESULT

A message says **Dismissed. This product will not be recommended again for 7 days.** (7 is the default; see System settings.)

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

REC-07 The Decided tab records who decided what

Manager

Before you start: REC-04 to REC-06 are done.

1. Click the **Decided** tab.

EXPECTED RESULT

Each decision is listed with its status (accepted, changed quantity, dismissed), who decided, when, and the note.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

REC-08 Cancel an accepted order

Manager

Before you start: REC-05 is done.

1. On the Decided tab, on the changed-quantity card, click **Cancel order** and confirm.

EXPECTED RESULT

A message says **Order cancelled. It no longer counts as on order.** The On order count goes down.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

REC-09 On order clears when the goods arrive

Manager

Before you start: REC-04 is done; note the product and quantity.

1. Open that product and **Record restock** with the same quantity.

EXPECTED RESULT

Restock recorded. On order goes back to 0 and on hand goes up. The product is not recommended again straight away.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

REC-10 Recalculate

Manager

1. Click **Recalculate**.

EXPECTED RESULT

A message says **Refreshing the recommendations. This takes a moment.** After a few seconds the page reflects the latest stock.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

REC-11 Filter the advice

Manager

1. Search `soap`; choose a category; choose a risk level.

EXPECTED RESULT

The cards follow each filter straight away.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

REC-12 Overstock tab

Manager

1. Click **Overstock**.

EXPECTED RESULT

Products with far more stock than they are likely to sell soon, for information (no order suggested).

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

B13. Notifications and email

Notifications appear under the bell and, if you choose, by email. In this set-up every email goes to Mailpit (`http://localhost:8026`), never to a real inbox. **No email may ever contain a person's name or email address.**

NOT-01 The bell — Manager

Before you start: There are notifications (for example after FOR-01).

1. Click the bell at the top right.

> EXPECTED RESULT
> A list of recent notifications opens; unread ones have a violet dot. The red number on the bell is the unread count.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks _____

NOT-02 Open and mark as read — Manager

1. Click one notification. Then open the bell again and click **Mark all as read**.

> EXPECTED RESULT
> Clicking takes you to the right page (for example the forecast). After Mark all as read the red number disappears.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks _____

NOT-03 All notifications — Manager

1. Open the bell → **See all notifications**.

> EXPECTED RESULT
> A page lists every notification, newest first.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks _____

NOT-04 Choose what comes by email

Manager

1. Settings → **Notifications**. For **Critical stock alerts** switch email off and keep in-app on. Save.

EXPECTED RESULT

A message says **Notification settings saved.** After reloading, the choices are kept.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

NOT-05 The replenishment digest

Manager

Before you start: A forecast exists. Digest frequency is Daily (the default).

1. In a terminal run: `docker compose exec app php artisan notifications:digest`
2. Open Mailpit.

EXPECTED RESULT

The terminal says **Digest sent to 2 people.** Mailpit has a digest for the Owner and the Manager: counts by risk level, the most urgent products, and a button to the recommendations.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

NOT-06 Critical stock alert

Manager

1. After a forecast (or REC-10), open Mailpit.

EXPECTED RESULT

An email titled **Critical stock: … may run out** lists products that may run out, to the Owner and Manager only. Each product is alerted at most once a day.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

NOT-07 Import with problems

Manager

Before you start: SAL-09 is done.

1. Look at the bell and Mailpit.

> EXPECTED RESULT
> A notification that the sales import had problems, with a link to its error report. Only the person who uploaded it is told.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

NOT-08 No personal data in any email

Any role

Before you start: Several emails are in Mailpit.

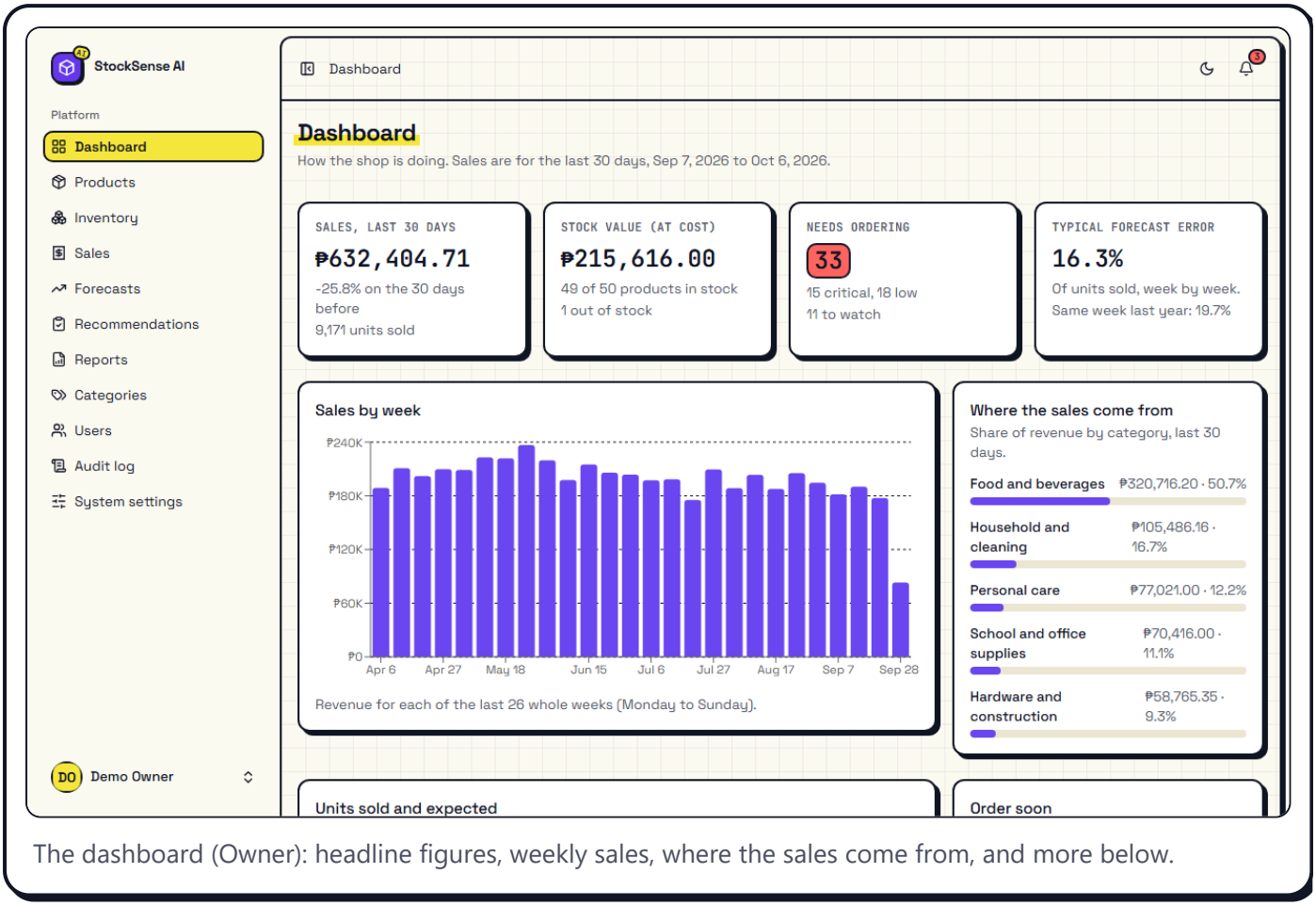
1. Open every email. Read the subject, the text and the links. Use Mailpit's **HTML** and **Text** tabs.

> EXPECTED RESULT
> None contains a person's name (Demo Owner, Demo Manager, a test user's name) or an email address. Greetings are just **Hello**. Emails are in the StockSense style.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

B14. Dashboard

The first page after signing in: how the shop is doing over the last 30 days compared with the 30 before.



The dashboard (Owner): headline figures, weekly sales, where the sales come from, and more below.

DSH-01 Headline figures

Owner

1. Open the **Dashboard**.

> EXPECTED RESULT
>
> Four cards: **Sales, last 30 days** (₱ and the change from the 30 days before), **Stock value (at cost)**, **Needs ordering** (a coloured number: red when anything is critical) and **Typical forecast error**.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

DSH-02 Cards lead somewhere

Owner

1. Click the **Needs ordering** card.

> EXPECTED RESULT
> The Recommendations page opens.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

DSH-03 Charts

Owner

1. Scroll through the page. Hover over the bars of **Sales by week**.

> EXPECTED RESULT
> Weekly sales for the last 26 weeks (a tooltip shows each week's figure), what is expected next with its range, revenue by category with shares, the most urgent products, and the best and slowest sellers.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

DSH-04 Figures agree with the rest of the app

Owner

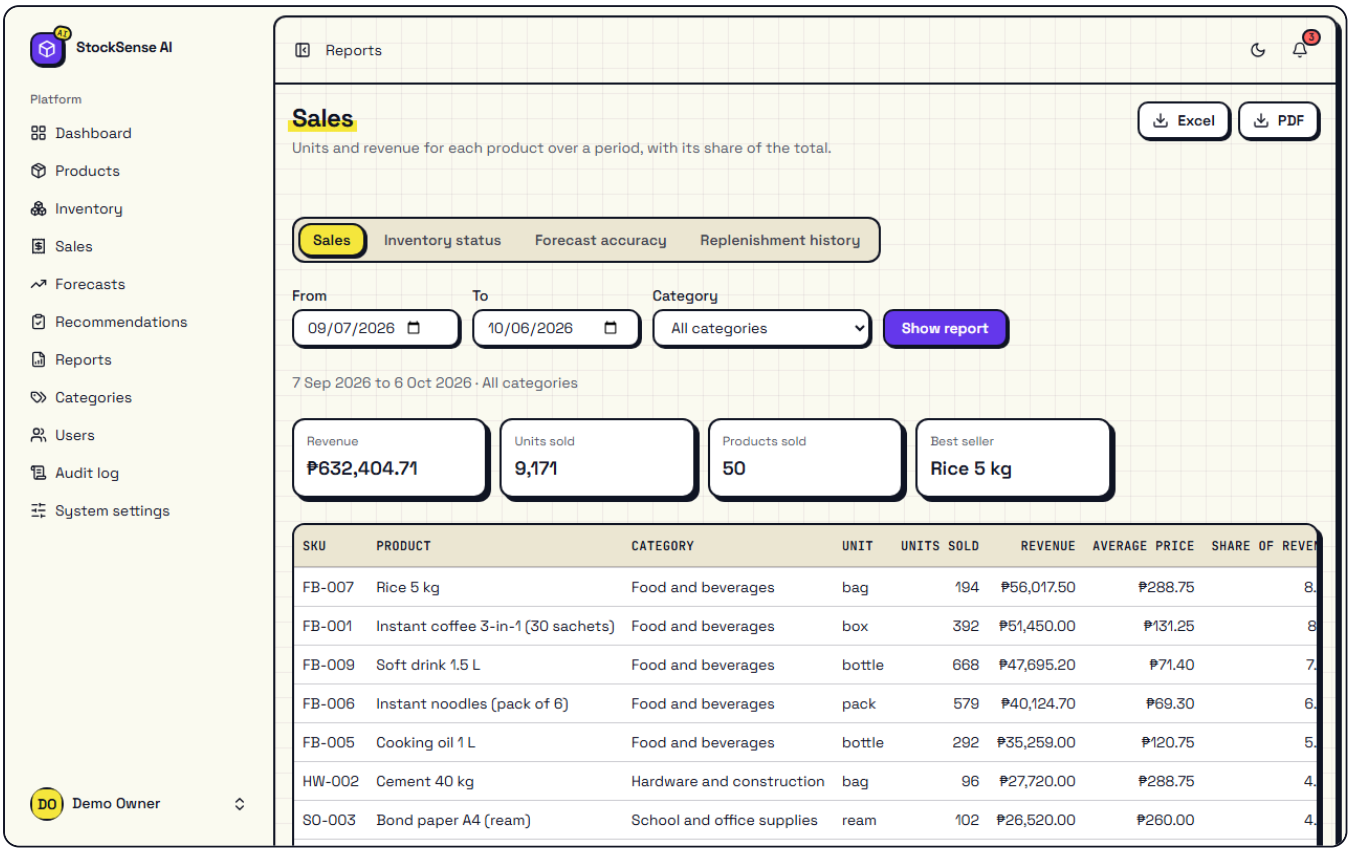
1. Compare the dashboard's **Needs ordering** number with the Critical + Low counts on Recommendations.

> EXPECTED RESULT
> They match.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

B15. Reports

Four reports, each on screen with filters, and downloadable as Excel (real numbers you can add up) or PDF (for printing).



The sales report with its filters, summary figures and the Excel and PDF buttons.

RPT-01 **The four reports** — **Owner**

1. Open **Reports**.

> EXPECTED RESULT
> Four cards: **Sales**, **Inventory status**, **Forecast accuracy** and **Replenishment history**.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks _____

RPT-02 Sales report with filters

Owner

1. Open **Sales**. Change From and To to last month and choose one category. Click **Show report**.

> EXPECTED RESULT
> The summary and the table show only that period and category. The address bar and the export buttons carry the same filters.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

RPT-03 Download as Excel

Owner

Before you start: RPT-02 is done.

1. Click the Excel button and open the file.

> EXPECTED RESULT
> A file named like `stocksense-sales-2026-09-01_2026-09-30.xlsx`. Numbers are real numbers (you can sum them), dates are dates, and an **About** sheet explains what the report covers.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

RPT-04 Download as PDF

Owner

1. Click the PDF button and open the file.

> EXPECTED RESULT
> A tidy A4 report: a violet StockSense tag, the title highlighted in yellow, summary boxes, a table with a dark header row and a yellow total row, and page numbers. Peso signs (₱) print correctly.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

RPT-05 The other three reports

Owner

1. Open Inventory status, Forecast accuracy and Replenishment history; export each once.

EXPECTED RESULT

Each opens with sensible figures for the demo shop and downloads in both formats.

☐ Pass ☐ Fail ☐ Blocked Remarks

RPT-06 Downloads are recorded

Owner

Before you start: RPT-03 and RPT-04 are done.

1. Open the **Audit log**.

EXPECTED RESULT

Entries say **Exported the sales report as an Excel file** and **… as a PDF**, with your name and the time.

☐ Pass ☐ Fail ☐ Blocked Remarks

B16. System settings

Owner only. How the advice is worked out and when things run automatically. Each setting shows its default; changes apply at once, including to background jobs.

SET-01 Change a setting — Owner

1. Open **System settings**. Set **Review period** to `14` days. Click **Save settings**.
2. Reload the page.

> EXPECTED RESULT
> A message says **Settings saved. They apply from now on.** After reloading it still says 14, with a note **Default: 7 days.**

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks _____

SET-02 Settings take effect — Owner

Before you start: SET-01 is done.

1. Open Recommendations and click Recalculate. Compare a card's numbers with before.

> EXPECTED RESULT
> Order-up-to quantities and safety stock are higher (ordering less often means holding more).

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks _____

SET-03 Out-of-range values are refused — Owner

1. Try **Review period** `0` and `61`; **Default service level** `100`; **Weekly forecast looks ahead** `27`.

> EXPECTED RESULT
> Each is refused with a red message giving the allowed range (1–60 days, 50–99.9%, 1–26 weeks).

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks _____

SET-04 The digest must come after the advice

Owner

1. Set **Digest emails are sent at** to a time earlier than **Recommendations are refreshed at**. Save.

> EXPECTED RESULT
> Refused: the digest must be after the recommendations are refreshed.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

SET-05 Saving without changes

Owner

1. Click **Save settings** without changing anything.

> EXPECTED RESULT
> A message says **Nothing was changed.**

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

SET-06 Restore defaults

Owner

Before you start: SET-01 is done.

1. Click **Restore defaults** and confirm in the dialog.

> EXPECTED RESULT
> A message says **The default settings are back.** Review period is 7 again.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

B17. Audit log

Owner only. Who did what and when, with what changed. It can be read but never edited.

AUD-01 Your actions are there — Owner

Before you start: You have done some of the tests above.

1. Open **Audit log**.

> EXPECTED RESULT
> Newest first: imports started and undone, forecasts started, recommendations accepted, settings changed, users created or deactivated, reports exported, each with the person and time.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks _____

AUD-02 What changed, from and to — Owner

Before you start: SET-01 or PRD-05 is done.

1. Find that entry.

> EXPECTED RESULT
> It shows the old and the new value (for example review period 7 → 14, or price 39.00 → 42.00).

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks _____

AUD-03 Filter the log — Owner

1. Filter by area, then by person, then by a period, then search `settings`.

> EXPECTED RESULT
> The list follows each filter.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks _____

AUD-04 **Nothing can be edited** **Owner**

1. Look for edit or delete buttons.

> **EXPECTED RESULT**
> There are none.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

B18. Security spot checks

Simple checks anyone can do with the browser. They do not replace a professional security test.

SEC-01 Protective headers

Signed out

1. Open the sign-in page. Press `F12` → **Network**, reload, click the first request (`login`) → **Headers** → Response headers.

EXPECTED RESULT

You see `X-Frame-Options: SAMEORIGIN`, `X-Content-Type-Options: nosniff`, `Referrer-Policy: strict-origin-when-cross-origin` and `Permissions-Policy: camera=(), microphone=(), …`. (The Content-Security-Policy header is switched on only in the production set-up, because it would block the development tools used here.)

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

SEC-02 Session cookies are protected

Signed out

1. `F12` → **Application** → **Cookies** → `http://localhost:8080`.

EXPECTED RESULT

The session cookie (its name ends in `-session`) is marked **HttpOnly**, so scripts on the page cannot read it.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

SEC-03 Typing in code does nothing

Manager

1. Create a category named `<script>alert(1)</script>`. Open the categories list. Delete the category afterwards.

EXPECTED RESULT

No pop-up appears; the name is shown as plain text.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks

SEC-04 Costly actions are slowed down when repeated

Manager

1. On **Recommendations**, click **Recalculate** 7 times within a minute.

> EXPECTED RESULT
> The first 6 work. The 7th shows **You are doing that too often. Please wait … and try again.** instead of an error page. After a minute it works again.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks _____

SEC-05 Blocked pages stay blocked

Inventory staff

1. As staff, try `/users/1/edit` and `/horizon`.

> EXPECTED RESULT
> Both are refused (403).

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks _____

B19. Speed, devices and browsers

How quick and how portable it feels. Write your measurements in the Remarks.

PRF-01 Page speed — Owner

Before you start: The app has been open for a while (not the first page after a start).

1. F12 → **Network**. Open Dashboard, Products, Recommendations and a report. Note the **Load** time at the bottom of the Network tab.

> EXPECTED RESULT
> Each page loads within about 2 to 3 seconds on a normal laptop. Write down the times.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks _____

PRF-02 Forecast time — Manager

1. Time a weekly forecast run from the click to the summary.

> EXPECTED RESULT
> Usually under a minute for the 50 products; up to a few minutes on a slower computer. Write down the time.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks _____

PRF-03 Other browsers — Any role

1. Open the app in Chrome, Edge and Firefox. Sign in and open a few pages in each.

> EXPECTED RESULT
> It looks and works the same in all three.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks _____

PRF-04 Phone-sized screen

Any role

1. `F12` → click the phone-and-tablet icon (device toolbar) → choose a phone (for example iPhone 12). Use the app.

> **EXPECTED RESULT**
> The left menu folds away behind a menu button. Pages fit the width without sideways scrolling. Tables scroll inside their box.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks _____

PRF-05 Keyboard only

Any role

1. Without the mouse, press `Tab` to move through the sign-in page and the dashboard, `Enter` to press buttons, `Esc` to close a dialog.

> **EXPECTED RESULT**
> You can reach and use every control. The focused control always has a clear violet outline. In a dialog, Tab stays inside it until you close it.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks _____

PRF-06 It recovers from a restart

Any role

1. In a terminal: `docker compose restart`. Wait a minute, then reload the app.

> **EXPECTED RESULT**
> You are still signed in or can sign in again. Your data (sales you entered, decisions) is all still there.

☐ **Pass** ☐ **Fail** ☐ **Blocked** Remarks _____

Part C · Rating sheet, bug reports, sign-off

When you have finished the test cases, rate the app against the eight quality characteristics of **ISO/IEC 25010**, the international standard for software product quality. Rate each statement from **1** (strongly disagree) to **5** (strongly agree): circle or tick one number. Base your rating on what you saw during testing.

> **Privacy.** Use the **Tester ID** your instructor gives you instead of your name. The sheet asks for no personal data (in line with the Data Privacy Act of 2012, RA 10173).

TESTER ID _____

DATE TESTED _____

COMPUTER _____

Computer: Windows version, RAM, and whether you used the Ubuntu folder or a Windows folder.

CHARACTERISTIC AND STATEMENT	1 2 3 4 5
1. Functional suitability: does it do what it should, correctly?	
The app has every function needed to forecast sales and plan restocking for a small shop.	(1) (2) (3) (4) (5)
Calculations and figures (stock, totals, forecasts, quantities to order) were correct where I could check them.	(1) (2) (3) (4) (5)
The functions help me reach the goal (knowing what to reorder, how much and when) without unnecessary steps.	(1) (2) (3) (4) (5)
The recommendations and their explanations made sense for the stock and sales shown.	(1) (2) (3) (4) (5)
2. Performance efficiency: is it fast and light enough?	
Pages opened quickly enough for everyday use.	(1) (2) (3) (4) (5)
A forecast run finished in an acceptable time, and I could keep working meanwhile.	(1) (2) (3) (4) (5)
Imports and report downloads were quick.	(1) (2) (3) (4) (5)

CHARACTERISTIC AND STATEMENT	1 2 3 4 5
3. Compatibility: does it work alongside other things?	
The app worked the same in the browsers I tried.	1 2 3 4 5
Its files (CSV, Excel, PDF) opened correctly in the programs I normally use.	1 2 3 4 5
It ran on my computer alongside my other programs without problems.	1 2 3 4 5
4. Usability: is it easy to learn and pleasant to use?	
I could find my way around without help.	1 2 3 4 5
Labels, messages and explanations were clear.	1 2 3 4 5
The app prevented mistakes or explained clearly how to fix them (for example in imports).	1 2 3 4 5
The look (light and dark mode) was attractive and easy to read.	1 2 3 4 5
It could be used with the keyboard alone and on a phone-sized screen.	1 2 3 4 5
5. Reliability: does it keep working?	
Nothing crashed or showed an error page during my tests.	1 2 3 4 5
After a restart, all my data was still there.	1 2 3 4 5
Bad input (wrong files, wrong values) was handled without breaking anything.	1 2 3 4 5
6. Security: is data protected?	
Each role could reach only what it should, even by typing addresses.	1 2 3 4 5
Passwords were handled safely (no password ever emailed; lockout after wrong tries).	1 2 3 4 5
Emails contained no personal data.	1 2 3 4 5

CHARACTERISTIC AND STATEMENT	1 2 3 4 5
Important actions were recorded in the audit log with who did them.	1 2 3 4 5
7. Maintainability: could it be changed and fixed easily? (Look at the README, `docs/` and the project layout.)	
The project is clearly organised and documented.	1 2 3 4 5
Error messages and logs helped me understand problems.	1 2 3 4 5
The app is made of clear parts (web app, forecasting service, database) that can be looked at separately.	1 2 3 4 5
8. Portability: can it be installed and moved easily?	
Installing it by following Part A was straightforward.	1 2 3 4 5
It could be reset, stopped and started again easily.	1 2 3 4 5
It would be easy to install on another computer.	1 2 3 4 5

Comments

What did you like most? What would you change first?

Bug report form

Fill in one for each failed test or problem you notice. Copy the form as many times as you need. Number them **BUG-1**, **BUG-2**, … and write the number in the test case's Remarks.

BUG NUMBER _____

TEST CASE ID _____

PAGE (ADDRESS) _____

SIGNED IN AS	_____
STEPS	_____
	_____
EXPECTED	_____
WHAT HAPPENED	_____
	_____
SEVERITY	☐ Critical (data lost or wrong, cannot continue) ☐ Major (a function does not work) ☐ Minor (works, but wrong or confusing) ☐ Cosmetic (looks)
SCREENSHOT	☐ Attached (press `Win` + `Shift` + `S` to take one) Light / dark mode: _____

Test summary and sign-off

MODULE	CASES	PASS	FAIL	BLOCKED
B1. Smoke test (5 minutes)	6			
B2. The front page	5			
B3. Signing in and your account	15			
B4. Light and dark mode	5			
B5. Roles and permissions	9			
B6. Users	10			
B7. Categories	6			
B8. Products	9			
B9. Stock and inventory	6			
B10. Sales and imports	12			
B11. Forecasts	9			
B12. Recommendations	12			
B13. Notifications and email	8			

MODULE	CASES	PASS	FAIL	BLOCKED
B14. Dashboard	4			
B15. Reports	6			
B16. System settings	6			
B17. Audit log	4			
B18. Security spot checks	5			
B19. Speed, devices and browsers	6			
Total	**143**			

TESTER ID _____

BUGS REPORTED _____

OVERALL VERDICT ☐ Ready to use ☐ Ready with minor fixes ☐ Not ready

DATE _____

Appendices

1. Demo accounts and addresses

These exist only in your local copy, for testing. They are never created on a real (production) installation.

EMAIL	PASSWORD	ROLE
`owner@stocksense.test`	`password`	**Owner**
`manager@stocksense.test`	`password`	**Manager**
`staff@stocksense.test`	`password`	**Inventory staff**

WHAT	ADDRESS
StockSense AI	`http://localhost:8080`
Mailpit (every email the app sends)	`http://localhost:8026`
Forecasting service API docs	`http://localhost:8001/docs`
Background jobs dashboard (Owner only)	`http://localhost:8080/horizon`
Database, for database tools (optional)	`localhost:5433`, database and user `stocksense`, password `secret`

2. Sample files

The test cases give small files to copy into Notepad (save with **Save as type: All files** and a name ending in `.csv`). Bigger ones come with the project:

FILE	WHAT IT IS
`web/database/data/sales-sample-with-errors.csv`	Sales with deliberate mistakes (unknown SKU, impossible date, a word as quantity)
`e2e/fixtures/sales-import.csv`	Five good sales rows
`e2e/fixtures/sales-with-problems.csv`	Three rows, two of them bad

FILE	WHAT IT IS
`web/database/data/products.csv`	The demo shop's 50 products, in the import layout
Download an example file button	On each import page: an empty file with the right columns

In the Ubuntu set-up, the project folder is reachable from Windows Explorer at `\\wsl$\Ubuntu\home\<your username>\stocksense-ai`.

3. Commands

Run them in the `stocksense-ai` folder.

TO	RUN
Start (after the first time)	`docker compose start`
Stop, keeping the data	`docker compose stop`
See what is running	`docker compose ps`
Reset to the clean demo shop	`docker compose exec app php artisan migrate:fresh --seed`
Send the digest email now	`docker compose exec app php artisan notifications:digest`
Make a weekly forecast in the terminal	`docker compose exec app php artisan forecast:run week --sync`
Refresh the recommendations	`docker compose exec app php artisan recommendations:generate --sync`
See a service's log	`docker compose logs horizon --tail 50` (or `app`, `ml`, `vite`)
Restart one service	`docker compose restart horizon`
Get the latest version of the code	`git pull` then `docker compose up -d --build` and `docker compose exec app php artisan migrate`
Remove everything, including the data	`docker compose down -v`
Run the automated tests (optional)	`docker compose exec app php artisan test`

4. Words used

WORD	MEANING
SKU	Stock-keeping unit: the shop's own code for a product, such as `FB-006`
Lead time	Days from placing an order to receiving it
Review period	How often the shop places orders (default 7 days)
Service level	How often you want to have stock when a customer asks, for example 95%
Safety stock	Extra stock to cover surprises while an order is on its way
Reorder point	The stock level at which to order: expected sales during the lead time plus safety stock
On order	Quantities you accepted that have not arrived yet
Days of cover	How many days the stock on hand would last at the usual rate of sales
Forecast range	The band around a forecast where actual sales should fall about 8 times in 10
Low confidence	A product with under a year of sales history, forecast with a plain recent average
MAE	Mean absolute error: the average number of units a forecast was off by
RMSE	Root mean squared error: like MAE, but big misses count more
MAPE	Mean absolute percentage error: the average miss as a percentage of actual sales
WAPE	Weighted absolute percentage error: all misses as a percentage of all units sold. The headline accuracy figure
Baseline	A simple guess the model must beat: the same week last year, or a recent average
Container	One part of the app running in Docker, with everything it needs

5. Where to read more

- `README.md`: the project overview and everyday commands
- `docs/user-guide.md`: how to use the app, page by page
- `docs/testing.md`: the automated tests and their results
- `docs/ml-methodology.md` and `docs/replenishment-methodology.md`: how the forecasts and the advice are worked out